

R. Cointe
Research Assistant;
currently, Institut Français
du Pétrole,
Rueil Malmaison, France

J.-L. Armand
Professor.
Department of Mechanical and Environmental
Engineering,
University of California,
Santa Barbara, Calif. 93106

Hydrodynamic Impact Analysis of a Cylinder

The problem of the vertical entry of a rigid horizontal cylinder into an incompressible inviscid fluid initially at rest is addressed. The contributions of previous researchers are presented and discussed in the light of various assumptions introduced and the validity of the results obtained. Based on this review, realistic simplifying assumptions are introduced and the problem formulated. The method of matched asymptotic expansions is used to solve the resulting boundary-value problem. A new formula for the impact force is obtained, which differs from the classical von Kármán's formula by a corrective term. The results obtained are compared with those of experimental observations and numerical calculations. The method may be extended to different geometries and nonvertical velocities to provide an estimate of the impact forces on the partially emerged pontoons of damaged semi-submersibles.

Introduction

Awareness of the current lack of knowledge concerning the various mechanisms affecting the stability of damaged semi-submersibles has stimulated worldwide research into this topic, both theoretical and experimental. The Norwegian Maritime Directorate initiated the Mobile Platform Stability (or MOPS) project in 1981. An overview of the project together with some recommendations are found in Dahle [1]. Another very comprehensive research project was launched by the Shipbuilding Research Association of Japan in 1982 to investigate stability of twin-pontoon semi-submersibles, with the preliminary results reported in [2]. Experimental results clearly demonstrate the importance of nonlinearities in the equations of motion when the pontoons pierce the free surface (Huang, Naess, and Hoff [3]). Obviously, the realistic prediction of the dynamic behavior of a semi-submersible in waves, wind and current represents a formidable challenge, as even the nature and the importance of the different phenomena involved are not known. Attempts at modeling have up until now necessarily been restricted in scope.

As a part of the MOPS project, Huang and Naess [4] and Naess and Hoff [5] developed a numerical simulation technique for heavily listed semi-submersibles. This numerical method was built based on the general method given by Paulling [6]. The main difficulty was in the determination of the forces acting on the partially emerged pontoons. Huang, Naess, and Hoff used a strip theory, and split the forces acting on a section into various components, representing a mix between rigorous expressions derived from hydrodynamic theory (neglecting the frequency dependence of the hydrodynamic coefficients) and semi-empirical formulas of the Morison type. The discrepancies reported between measured and calculated responses could very well be explained by these uncertainties on the loading terms, which should be determined with higher accuracy.

The need for a rational approach to the problem of the determination of the wave and current forces acting on the inclined pontoons is apparent. As Lin, Newman, and Yue [7] point out, when viscous forces are neglected, the two main difficulties to overcome in a situation of this kind are related to the memory effect of the free surface (i.e., the presence of outgoing waves), and the mathematical treatment of the flow near the intersection of the body and the free surface. These problems must be addressed in order to deal with the nonlinearities introduced by the fact that the pontoons are piercing the free surface, nonlinearities, which apparently play an essential role.

Insight into this problem is gained from the consideration of the simpler, yet not fully understood, situation of the hydrodynamic impact of a circular cylinder. This represents a first but meaningful step toward the determination of impact forces acting on the partially emerged pontoons, the importance of which may be considerable. Consideration of this problem should also lead to a better knowledge of the phenomena involved in the determination of the flow around the pontoons.

The Water Impact Problem

The problem of water impact was initially addressed in order to predict the impact force on a landing seaplane. In a pioneering paper, von Kármán [8] introduced the main physical features of the phenomenon. He confined himself to a study of the vertical entry of a wedge, but his results may also apply to the case of a circular cylinder. Von Kármán equated the impact force on the penetrating object to half the force acting on a flat plate moving in an unbounded fluid at the impacting velocity. This force is given by

$$F = \frac{d}{dt} (mV) \quad (1)$$

where m is half the added mass of the plate of width given by the intersection of the body and the undisturbed free surface.

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Wagner [9, 10] subsequently refined von Kármán's analysis and offered the first mathematical treatment of the problem. Linearizing it in the case of a wedge with small deadrise angle, he approximated the body by a flat plate and obtained a zero potential free surface boundary condition. He took into account the piled-up water to determine the wetted width, and verified von Kármán's formula (equation (1)) for the impact force. Yet he took for m half the added mass of the plate of width the *wetted* width of the wedge. This modification to von Kármán's formula will be referred to later as the *wetting correction*.

Fabula [11] used an ellipse-fitting method to calculate the wetting correction but considered the added mass of the exact shape in equation (1). This method was developed to avoid the singularity which appears in the linearized problem when flat plate-fitting is used.

Eventually, it was suggested that von Kármán's formula should be used with the added mass of the submerged body, the wetting correction, and also a *drag correction* taking into account the quadratic terms in the pressure and other effects. In particular, even for the case when the free surface boundary condition is taken as $\Phi = 0$ on the undisturbed free surface, Schiffman and Spencer [12] pointed out that the exact potential theory force is not given by equation (1), but has to be modified because the body is not fully wetted.²

Experimental as well as theoretical studies of the wave slam on horizontal members of cylindrical cross sections were recently completed. Faltinsen et al. [14] applied von Kármán's formula using for the value of the added mass that of a circular lens, but did not consider the wetting and drag corrections.

Comprehensive numerical and experimental results concerning the water impact of a circular cylinder were given in an EPRI report [15] as a part of research conducted into the safety of boiling water reactors. The four numerical models

presented in this report are:

- 1 an explicit Lagrangian method using the STEALTH code (Gross [15]);
- 2 a boundary integral method based on a discretization of the free surface (Geers [15]);
- 3 a finite element method (Marcal [15]);
- 4 an incompressible Eulerian fluid method using the SOLA-SURF code (Nichols and Hirt [15]).

The numerical results obtained, together with experimental results and those obtained by applying some of the analytical formulas described in the foregoing, are shown on Fig. 1. The

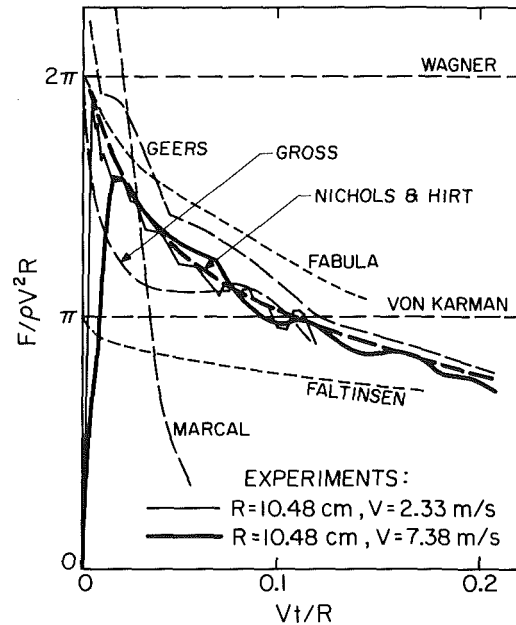


Fig. 1 Experimental, numerical, and theoretical slamming coefficients

Nomenclature

B = body boundary (equation (3))
 F = impact force
 F_B, F_I, F_Q, F_V = contributions to the impact force
 Fr = Froude number (equation (5))
 H = penetration depth
 i = square root of -1
 l = half-wetted width
 L = length scale (equation (5))
 m = added mass
 o = much smaller than
 p = pressure
 R = cylinder radius
 S = parameterization of free surface (equation (2))
 t = time
 u = intermediate variable (equations (18) and (19))
 v = time derivative of l

V = impacting velocity
 w = complex potential associated with φ
 W = complex potential associated with Φ
 x = horizontal coordinate
 z = vertical coordinate
 δ = jet thickness
 Δ = Laplacian (∇^2)
 ϵ = perturbation parameter (equation (5))
 ζ = complex variable, $x + iz$
 η = free surface elevation (equation (8))
 ξ = height of piled-up water
 ρ = density of water
 φ = velocity potential (equation (16))
 Φ = velocity potential
 ∇ = gradient

Subscripts

i = i th order approximation ($i = 0, 1, 2$)
 s = stagnation point
 t = partial derivative with respect to t
 T = Taylor's solution [17]
 x = partial derivative with respect to x
 z = partial derivative with respect to z

Other Symbols¹

$\tilde{a}$ = nondimensional variable (equation (7))
 q = outer domain variable (equation (9))
 a^* = inner domain variable (equation (14))

¹ These symbols are not used on subscripts indicating differentiation. Except when otherwise specified, derivatives are taken with respect to the current variables.

nondimensional impact force (often referred to as the *slamming coefficient*) is plotted as a function of the nondimensional time. The main conclusions which may be drawn from this figure are as follows:

(a) Von Kármán's formula underestimates the impact force, while Wagner's formula overestimates it.

(b) The use of the added mass of the circular lens in von Kármán's formula does not improve its accuracy. On the other hand, Fabula's method appears to be very accurate.

(c) For all the analytical results given, the impact force rises instantaneously from zero to a finite value. This is not in agreement with the experimental data. Several factors, specifically the formation of an air cushion between the cylinder and water surface, free-surface deformation prior to the impact, entrapped gases in the water, and surface irregularities, might account for some finite rise time. The assumptions introduced in a simple inviscid flow approach cannot lead to a correct solution to the problem at the very beginning of impact.

(d) The compressibility of the water should also be considered at the beginning of impact, when the apparent growth velocity of the circular lens is greater than the sound velocity in the liquid. This effect, however, appears to be negligible in comparison to the phenomena described in (c).

(e) The results of three of the numerical simulations are in good agreement with the experimental data. The effects described in (c) and (d) being neglected in these simulations, as well as the flexibility of the cylinder, it seems legitimate to take advantage in the analysis which follows of the same simplifications. Even more important, the fact that the experimental results, scaled without consideration for these effects, are almost superimposed, except during the rise time, justifies these simplifications.

In conclusion, based on the set of data shown, it appears that an accurate prediction of the impact force can be made under the general simplifying assumptions stated by Wagner (incompressible and inviscid fluid, irrotational flow, rigid cylinder, no surface tension, and eventually no gravity), except for the very beginning of impact where other effects should be considered. Those are the assumptions which will be used in what follows.

Water Impact of a Circular Cylinder

Statement of the Problem. The penetration at constant vertical velocity $-V$ ($V > 0$) of an infinite circular cylinder of radius R into a fluid domain initially at rest is considered here. The undisturbed free surface coincides with the x -axis of the fixed (x, z) coordinate system. It is assumed that the cylinder is rigid and that the fluid is incompressible and inviscid. All the quantities involved may therefore be expressed in terms of the velocity potential, which will be chosen as the main unknown variable. The surface tension of the water is neglected. The equation of the free surface is given in implicit form by

$$S(x, z, t) = 0 \quad (2)$$

If the origin of time is chosen to be the instant of impact, the body boundary is given for $Vt < R$ by

$$z = B(x, t) = R - \sqrt{R^2 - x^2} - Vt \quad (3)$$

Because of the symmetry of the problem, only the domain $x \geq 0$ is considered. The different boundaries are represented on Fig. 2. With these definitions and assumptions, the equations of the problem are easily derived. The problem may be stated as follows:

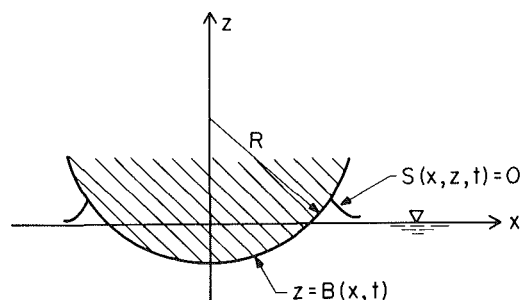


Fig. 2 Geometric definitions

Field equation:

$$\Delta \Phi = 0 \quad \text{in the fluid domain} \quad (4a)$$

Initial conditions:

$$\Phi(x, z, 0) = 0; \quad S(x, 0, 0) = 0 \quad (4b)$$

Kinematic and dynamic free-surface boundary conditions

$$\left. \begin{aligned} S_t + \nabla S \cdot \nabla \Phi &= 0 \\ gz + \Phi_t + \frac{1}{2} \nabla \Phi \cdot \nabla \Phi &= 0 \end{aligned} \right\} \quad \text{for } S(x, z, t) = 0 \quad (4c)$$

Body boundary condition

$$\Phi_{,x} - \Phi_{,z}(R - Vt - z) = V(R - Vt - z) \quad \text{for } z = B(x, t) \quad (4d)$$

Symmetry condition

$$\Phi_{,x} = 0 \quad \text{for } x = 0, \quad z \leq -Vt \quad (4e)$$

Conditions at infinity

$$\nabla \Phi \rightarrow 0 \quad \text{and} \quad \Phi_t \rightarrow 0 \quad \text{for } (x^2 + z^2) \rightarrow \infty \quad (4f)$$

Even with the simplifying assumptions introduced, two fundamental nonlinearities remain. The first is related to the fact that the free surface boundary conditions are given on a surface which itself is to be found as a part of the problem. The second arises from nonlinear terms in the boundary conditions themselves. A perturbation method will be used to find an approximate solution to this boundary-value problem. In order to perform the appropriate simplifications, nondimensional variables are first defined.

Nondimensional Variables. Let H be the penetration depth. The wetted width of the cylinder is proportional to $\sqrt{HR}$ at the beginning of impact. The length scale L is chosen to scale the wetted width, and the following quantities are introduced:

$$L = \sqrt{HR}, \quad \epsilon = \frac{H}{L} = \sqrt{\frac{H}{R}}, \quad \text{and} \quad \text{Fr} = \frac{V}{\sqrt{gR}} \quad (5)$$

The velocity potential should be scaled by VL and the linear term in Bernoulli's equation, Φ_t , should be scaled by V^2L/H . The impact force resulting from this contribution to the pressure should therefore be proportional to ρV^2R and be a constant, in agreement with von Kármán's and Wagner's results. The quadratic term in Bernoulli's equation, $\nabla \Phi \cdot \nabla \Phi$, should be scaled by V^2 and the quadratic correction of the preceding term should be proportional to $\epsilon \rho V^2R$ and vary like the square root of the penetration depth (or time). If gravity is neglected, we are therefore led to consider a perturbation method where the small parameter is ϵ . It will, therefore, be assumed that

$$\epsilon \ll 1 \quad \text{and} \quad \epsilon \ll \text{Fr} \quad (6)$$

The nondimensional variables are then defined by

$$\bar{\Phi} = \frac{\Phi}{VR}, \quad \bar{x} = \frac{x}{L}, \quad \bar{z} = \frac{z}{L},$$

$$\bar{t} = \frac{Vt}{H}, \quad \bar{S} = \frac{S}{L}, \quad \bar{B} = \frac{B}{L} \quad (7)$$

The velocity potential has been scaled by the product of the velocity and the cylinder radius (which is the only length scale of the problem independent of time) and is, therefore, expected to be of order ϵ .

Outer Domain Equations. In addition, it is assumed that in an outer domain far from the water line the equation of the free surface is given by

$$\bar{S}(\bar{x}, \bar{z}, \bar{t}) = \bar{z} - \bar{\eta}(\bar{x}, \bar{t}) = 0 \quad (8)$$

The intersection of the free surface and the body (or water line) being the point $(\bar{l}(\bar{t}), \bar{\xi}(\bar{t}))$, a change of variable is made to fix its position at point $(1, 0)$. The following new variables are, therefore, introduced:

$$\bar{x} = \frac{\bar{x}}{\bar{l}}, \quad \bar{z} = \frac{\bar{z} - \bar{\xi}}{\bar{l}}, \quad \bar{\Phi} = \frac{\bar{\Phi}}{\bar{l}},$$

$$\bar{\eta} = \frac{\bar{\eta} - \bar{\xi}}{\bar{l}^2}, \quad \bar{B} = \frac{\bar{B} - \bar{\xi}}{\bar{l}}, \quad \bar{t} = \bar{t} \quad (9)$$

We assume for the unknowns the following expansions:

$$\bar{\Phi} = \epsilon \bar{\Phi}_1 + o(\epsilon), \quad \bar{\eta} = \epsilon \bar{\eta}_1 + o(\epsilon) \quad (10a)$$

$$\bar{l} = \bar{l}_1 + o(1), \quad \frac{d\bar{l}}{d\bar{t}} = \bar{v} = \bar{v}_1 + o(1)$$

$$\bar{\xi} = \epsilon \bar{\xi}_1 + o(\epsilon) \quad (10b)$$

The first-order outer domain problem is now easily found. Equation $\bar{B}(1, 0) = 0$ yields at this order $\bar{l}_1^2 = 2(\bar{t} + \bar{\xi}_1)$ and equations (4a) to (4f) become

$$\Delta \bar{\Phi}_1 = 0 \quad (11a)$$

$$\bar{l}_1(0) = 0 \quad (11b)$$

$$\left. \begin{aligned} 2\bar{l}_1 \bar{v}_1 \bar{\eta}_1 + \bar{l}_1 \bar{v}_1 - 1 + \bar{l}_1^2 \bar{\eta}_{1t} - \bar{l}_1 \bar{v}_1 x \bar{\eta}_{1x} - \bar{\Phi}_{1z} &= 0 \\ \bar{v}_1 \bar{\Phi}_1 + \bar{l}_1 \bar{\Phi}_{1t} - \bar{v}_1 x \bar{\Phi}_{1x} &= 0 \end{aligned} \right\}$$

$$\text{for } \bar{z} = 0, \quad \bar{x} \geq 1 \quad (11c)$$

$$\bar{\Phi}_{1z} = -1 \quad \text{for } \bar{z} = 0, \quad \bar{x} \leq 1 \quad (11d)$$

$$\bar{\Phi}_{1x} = 0 \quad \text{for } \bar{x} = 0, \quad \bar{z} \leq 0 \quad (11e)$$

$$\nabla \bar{\Phi}_1 \rightarrow 0 \quad \text{and} \quad \left(\frac{\partial \bar{\Phi}_1}{\partial \bar{t}} \right)_1 \rightarrow 0 \quad \text{for } (\bar{x}^2 + \bar{z}^2) \rightarrow \infty \quad (11f)$$

We shall now show that there exists a solution of equations (11) which is independent of $\bar{t}$. This is equivalent to saying that, at first order, the problem admits a self-similar solution. Assuming that such a solution exists and making use of equation (11f), the dynamic free surface boundary condition reduces to

$$\bar{\Phi}_1 = 0 \quad \text{for } \bar{z} = 0, \quad \bar{x} \geq 1 \quad (12)$$

$\bar{\Phi}_1$ is therefore defined as the potential of the unbounded flow around a flat plate of half width $\bar{l}$. Going back to the overlined variables, $\bar{\Phi}_1$ is the unbounded flow around a flat plate of half width $\bar{l}_1$. If this length were half the width of the circular lens defined by the cylinder and the $\bar{x}$ -axis, this would correspond to von Kármán's solution for the flow [8]. But this length represents in fact half the wetted width of the cylinder and corresponds to Wagner's solution [9, 10], where the piled-up

water is taken into account. We have therefore justified the wetting correction which appears at the first order in the linearized problem.

Since the flow around a flat plate is easily obtained using conformal mapping techniques, $\bar{\Phi}_1$ is known. Still, in order to find $\bar{\Phi}_1$, $\bar{l}_1$ and the free surface elevation have to be found. This can be accomplished using the kinematic free surface boundary condition. In order to obtain a solution independent of time, $\bar{l}_1 \bar{v}_1$ has to be a constant. Using the fact that $\bar{\eta}_1(1, \bar{t}) = 0$, and imposing the condition that $\bar{\eta}_1$ be bounded at infinity, a solution of equations (11) is completely determined:

$$\bar{l}_1 = 2\sqrt{\bar{t}}, \quad \bar{\xi}_1 = \bar{t}, \quad \bar{\eta}_1 = -\frac{1}{2} - \frac{x\sqrt{x^2-1}}{2} + \frac{x^2}{2} \quad (13)$$

The height of the piled-up water, $\bar{\xi}_1$, is exactly equal to the penetration depth. This explains the importance of the wetting correction in this case.

Figure 3 compares the analytical result for the free surface elevation with that of a numerical calculation using a discretization of the fluid domain by finite differences (Nichols and Hirt [13]). The agreement is excellent for $V\bar{l}$ much smaller than R . In order to improve the accuracy of the calculation for larger values of time, one should go to the next order approximation since ϵ^2 is essentially the nondimensional time. Yet a major problem arises because the first-order solution, given by the flow around a flat plate, is singular at the edges of the plate. The fact that $\bar{\Phi}_1$ is not bounded near the water line is in contradiction with the assumptions made to linearize the problem. The outer solution is therefore not valid near the intersection of the body and the free surface. Near the singularity, nonlinearities have to be taken into account and a new scaling performed. This will now be done using the method of matched asymptotic expansions in order to define the nonlinear inner problem and solve it.

Inner Domain Equations. The inner variables are defined in the vicinity of the singularity and scaled by H^2/L . This choice of the inner length scale appears to be the only one possible in order to match the inner and outer solutions.

$$x^* = \frac{1}{\epsilon^2}(x-1), \quad z^* = \frac{1}{\epsilon^2}z, \quad t^* = \bar{t} \quad (14a)$$

$$\Phi^* = \frac{1}{\epsilon^2}\bar{\Phi}, \quad S^* = \frac{1}{\epsilon^2}\bar{S} = \frac{1}{\epsilon^2}\bar{S} \quad (14b)$$

Only the zeroth-order inner problem will be considered here, where

$$\Phi^* = \Phi_0^* + o(1), \quad S^* = S_0^* + o(1) \quad (15)$$

With the change of variable

$$\varphi^* = \Phi_0^* - \bar{v}_1 x^* \quad (16)$$

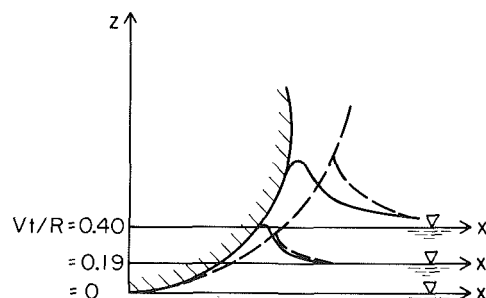


Fig. 3 Free surface elevation: numerical (—) and first-order analytical (---) solutions (the dotted parabola represents the first-order approximation of the body boundary)

the zeroth-order inner problem is stated as

$$\Delta \varphi^* = 0 \quad (17a)$$

$$\left. \begin{aligned} \nabla \varphi^* \cdot \nabla S_0^* &= 0 \\ \nabla \varphi^* \cdot \nabla \varphi^* &= \bar{v}_1^2 \end{aligned} \right\} \text{ for } S_0^*(x^*, z^*, t^*) = 0 \quad (17b)$$

$$\varphi_z^* = 0 \text{ for } z^* = 0 \quad (17c)$$

This φ^* -problem is a classical jet problem. The normal velocity is zero on the x^* -axis, defining the body boundary. The velocity is tangential and of constant magnitude $\bar{v}_1$ on the free surface. A solution can be found using the Schwarz-Christoffel method. In terms of an intermediate variable u , the complex potential, w^* , and the complex variable, ζ^* , are expressed as

$$w^* = -\frac{2}{\pi} \bar{v}_1 \delta^* \left(\frac{u^2}{2} + \log u + \frac{1}{2} + i \frac{\pi}{2} \right) \quad (18)$$

$$\zeta^* - \zeta_s^* = \frac{2}{\pi} \delta^* \left(\frac{u^2}{2} - 2iu - \log u + \frac{5}{2} - i \frac{\pi}{2} \right) \quad (19)$$

This completely defines the velocity field in the inner domain, provided that δ^* , the thickness of the jet, and ζ_s^* , the position of the stagnation point, are known (Fig. 4). The inner and outer solutions have to be matched in order to determine these constants.

Matching of the Inner and Outer Solutions. The matching is performed by requiring that the outer solution near the singularity and the inner solution at infinity have the same behavior. It is easy to show that matching of the potentials and the free surfaces is realized for

$$\delta^* = \frac{\pi}{8 \bar{v}_1^2} = \frac{\pi}{8} t^* \quad (20)$$

This is true for any value of ζ_s^* . The inner problem is not fully determined by the classical matching procedure. ζ_s^* will be found later (see equation (37)) by matching of the loci of the maximum pressure. Apparently, Wagner [10] must have assumed for the wedge $\zeta_s^* = 0$. He obtained a similar formula for the thickness of the jet. Expressed in terms of the dimensional variables, this thickness is

$$\delta = \frac{\pi R}{4} \left(\frac{Vt}{R} \right)^{3/2} \quad (21)$$

In the numerical model given by Nichols and Hirt [15], no jet appears (Fig. 3). This is due to the procedure used to get a smooth transition between the body and the free surface boundary conditions which essentially consists in applying a pressure on the free surface before the actual contact with the body, which clearly does not allow for a jet to be created. Greenhow and Lin [16] did photographic studies of the water impact of a cylinder where a jet appeared at each water line, in qualitative agreement with the preceding results.

Since it was possible to remove the singularity encountered in the outer approximation, the first-order impact force may now be computed.

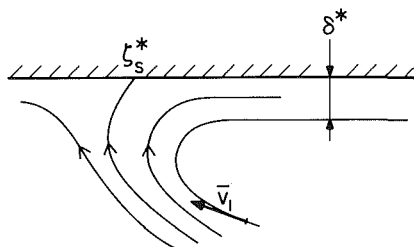


Fig. 4 Inner domain solution (φ^* -problem)

Impact Force

First-Order Impact Force. In order to obtain the first-order impact force on the cylinder, the pressure on the body is first computed using Bernoulli's theorem. The nondimensional pressure is defined by

$$\bar{p} = \underline{p} = p^* = \frac{p}{\rho} \frac{T}{VR} \quad (22)$$

In the outer domain and at the first order, this leads to the following expression for the pressure on the body:

$$\underline{p}(x, B, t) \approx -\epsilon(\bar{v}_1 \Phi_1 - \bar{v}_1 x \Phi_{1,x}) = \frac{\epsilon \bar{v}_1}{\sqrt{1-x^2}} \quad (23)$$

In the inner domain, the pressure on the body at the zeroth order is given by

$$p^*(x^*, 0, t^*) = \frac{1}{2}(\bar{v}_1^2 - \varphi_x^{*2}) \quad (24)$$

An exact expression for this pressure can be found using equations (18) and (19). The maximum pressure is $\frac{1}{2}\bar{v}_1^2$, and is reached at the stagnation point. It can be shown that the contribution of the inner pressure to the impact force is of order ϵ^2 , the width of the inner domain. Thus the first-order impact force is simply given by integration of the first-order outer pressure, yielding

$$\frac{F}{\rho V^2 R} \approx 2\pi \quad (25)$$

This result was a priori not obvious since the contribution of the inner domain solution might have been of the first order. It corresponds to Wagner's modification [9, 10] of the original von Kármán's formula [8], and may be rewritten as

$$F = \frac{d}{dt}(mV) + o(\rho V^2 R) \text{ for } \frac{Vt}{R} \rightarrow 0 \quad (26)$$

where m is half the added mass of the flat plate of width the wetted width of the cylinder, $2l_1$. Therefore, von Kármán's formula, equation (1), when it is used with flat plate-fitting and the wetting correction, corresponds to the first-order approximation of the impact force.

The wetting correction multiplies the width of the flat plate by a factor $\sqrt{2}$, but the impact force by a factor 2. As was previously seen, this value of the force overestimates the experimental results, which all lie between Wagner's and von Kármán's approximations (Fig. 1). The fact that the agreement between the first-order prediction of the impact force and the experimental data deteriorates with time is not surprising because the perturbation parameter, ϵ , is proportional to the square root of the time scale. Due to the assumptions introduced, each successive approximation is independent of ϵ ; it is logical for the first one to be independent of time. In order to improve the accuracy, and to model the time dependence, the next order must be considered. For this purpose, it is now necessary to take into account:

- (a) the zeroth-order inner domain pressure, which is of order 1 on a domain of width ϵ^2 ;
- (b) the first-order outer domain quadratic pressure, which is of order ϵ^2 on a domain of order 1;
- (c) the second-order outer domain linear pressure, which is of order ϵ^2 on a domain of order 1.

An "exact" expression for the second-order impact force, therefore, requires the determination of ζ_s^* and the solution of the next order outer problem.

Second-Order Impact Force. The following expansions of

the unknowns are now assumed:

$$\Phi = \epsilon \Phi_1 + \epsilon^2 \Phi_2 + o(\epsilon^2), \quad \eta = \epsilon \eta_1 + \epsilon^2 \eta_2 + o(\epsilon^2) \quad (27a)$$

$$\bar{T} = \bar{T}_1 + \epsilon \bar{T}_2 + o(\epsilon), \quad \bar{v} = \bar{v}_1 + \epsilon \bar{v}_2 + o(\epsilon) \quad (27b)$$

$$\bar{\xi} = \epsilon \bar{\xi}_1 + \epsilon^2 \bar{\xi}_2 + o(\epsilon^2). \quad (27c)$$

Equation $B(1, 0) = 0$ yields at this order $\bar{\xi}_2 = \bar{T}_1 \bar{T}_2$. The second-order outer problem can be found using a Taylor expansion of the boundary conditions. It is possible to remove the time dependence of this problem, which reduces then to a Neumann-Dirichlet boundary-value problem. Still, we did not succeed in obtaining an analytical solution for it. An important simplification will be introduced at this stage to allow its solution, that is, the dynamic free surface boundary condition will be taken as

$$\Phi_2 = 0 \quad \text{for } z = 0, \quad x \geq 1 \quad (28)$$

The resulting boundary-value problem will now be solved using the solution for the unbounded flow around a circular lens given by Taylor [17]. This solution can be expanded in terms of ϵ , which characterizes the exterior angle of intersection of the two circles. It can be verified that a solution of the approximate second-order problem, where the dynamic free surface boundary condition is assumed given by equation (28), is given by

$$\Phi_2 = \lim_{\epsilon \rightarrow 0} \frac{\Phi_T - \Phi_1}{\epsilon} \quad (29)$$

Using this approximation for the second order potential, the kinematic free surface boundary condition can be used to obtain the free surface elevation in a manner similar to what was done for the first-order outer domain problem. This yields

$$\bar{T}_2 = -\frac{4}{9\pi} \bar{T}, \quad \bar{\xi}_2 = -\frac{8}{9\pi} \bar{T}^{3/2}, \quad (30a)$$

$$\eta_2 = -\frac{4}{3\pi} x \sqrt{x^2 - 1} \sqrt{\bar{T}} - \frac{1}{6\pi} (4x^2 - 1) \sqrt{x^2 - 1} \sqrt{\bar{T}} \log\left(\frac{x-1}{x+1}\right) \quad (30b)$$

The second-order solution being approximated, it is now possible to find the second-order impact force.

Its value is obtained by integrating the pressure distribution on the cylinder. The different contributions to this pressure will be considered separately in order to bring into light the effects involved in the drag correction. The added mass of the circular lens is given by

$$\begin{aligned} \bar{m} &= -2\epsilon \int_0^1 \bar{\Phi}(\bar{x}, \bar{B}, \bar{T}) d\bar{x} \\ &= -2\epsilon \bar{T}^2 \int_0^1 \Phi(x, B, \bar{T}) dx \end{aligned} \quad (31)$$

At the second order, and using Taylor's solution, this leads to

$$\bar{m} \approx \epsilon 2\pi \bar{T} - \epsilon^2 \frac{32}{9} \bar{T}^{3/2} \quad (32)$$

Direct application of von Kármán's formula, with the added mass of the circular lens and the wetting correction up to the second order, would give

$$\bar{F}_v = \frac{d}{dt} \bar{m} \approx \epsilon 2\pi - \epsilon^2 \frac{16}{3} \sqrt{\bar{T}} \quad (33)$$

Still, this correction to Wagner's result is not sufficient. Other

corrections, of the same order of magnitude, have also to be made. The first one stems from the need to take into account the relative motion of the cylinder and the free surface. Differentiating equation (31) yields

$$\frac{d}{dt} \bar{m} = -2\epsilon \int_0^1 [\bar{\Phi}_t + \bar{\Phi}_z \bar{B}_t](\bar{x}, \bar{B}, \bar{T}) d\bar{x} \quad (34)$$

The following term has therefore to be added to $\bar{F}_v$:

$$\bar{F}_B = 2\epsilon \int_0^1 [\bar{\Phi}_z \bar{B}_t](\bar{x}, \bar{B}, \bar{T}) d\bar{x} \approx 4\epsilon^2 \sqrt{\bar{T}} \quad (35)$$

The importance of this correction should not be underestimated. $\bar{F}_v + \bar{F}_B$, representing the integration of the linear pressure (up to the second order, and with the approximations made) over the whole cylinder, is much closer to 2π than $\bar{F}_v$.

Since the integration should only be performed over the outer domain, and the inner pressure integrated over the inner domain, a new correction has to be introduced:

$$\begin{aligned} \bar{F}_I &= 2\bar{T}_1 \epsilon^2 \int_{\text{inner domain}} p_0^* dx^* \\ &\quad - 2\bar{T}_1 \epsilon \int_{\text{inner domain}} p_1 dx \end{aligned} \quad (36)$$

Part of this correction stems from the fact that, at the first order, the integral of the linear pressure over the inner domain (of width $\approx \epsilon^2$) is of order ϵ^2 , since this pressure is not bounded. In order to compute $\bar{F}_I$, the bounds of the inner domain have to be found. It can be shown that, at the lowest order, maximum pressures in the inner and outer domains are the same. This leads to locate the stagnation point of the inner domain in that point, fixed in the inner domain, where the outer pressure is maximum and gives

$$x_s^* = -\frac{4}{\pi} \delta^* \quad (37)$$

Equation (37) may be considered as the matching condition for the inner and outer pressures. The outer domain now extends from the origin to the stagnation point, and the inner domain from the stagnation point to infinity. This gives for $\bar{F}_I$

$$\bar{F}_I = -2\epsilon^2 \sqrt{\bar{T}} \quad (38)$$

The first-order quadratic pressure should also be integrated over the outer domain only. Actually, it would yield an infinite suction force if integrated over the whole cylinder. This leads to the correction

$$\begin{aligned} \bar{F}_Q &= -\epsilon^2 \bar{T}_1 \int_{\text{outer domain}} \frac{1}{1-x^2} dx \\ &= \epsilon^2 \sqrt{\bar{T}} (\log(\epsilon^2 \bar{T}) - 2 \log 2) \end{aligned} \quad (39)$$

The total impact force is now

$$\begin{aligned} \bar{F} &= \bar{F}_v + \bar{F}_B + \bar{F}_I + \bar{F}_Q \\ &= \epsilon 2\pi - \epsilon^2 \sqrt{\bar{T}} (10/3 + 2 \log 2 - \log(\epsilon^2 \bar{T})) \end{aligned} \quad (40)$$

This formula for the impact force is compared on Fig. 5 with the experimental results already shown. The agreement is excellent for $\epsilon \bar{T}$ small, except for the finite rise time not predicted by this theory. The analytical result overestimates the impact force relatively rapidly. This is due to the fact that the perturbation parameter varies like the square root of the time scale, and also to the fact that this result is only an approximation of the second-order problem, and that a new correction, at this order, should be made to take into account

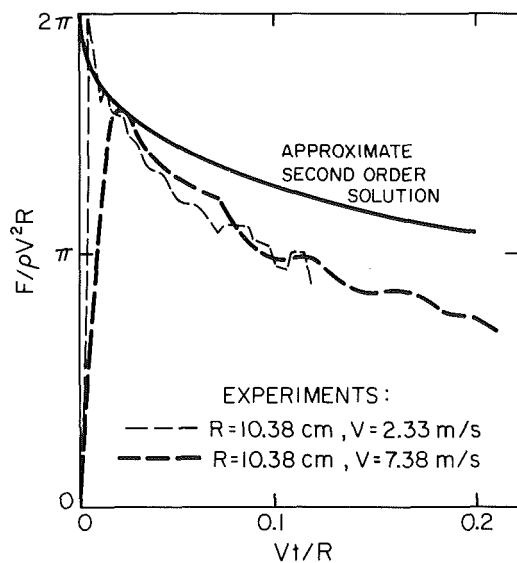


Fig. 5 Second-order and experimental slamming coefficients

the geometry of the free surface. The rapid decrease of the nondimensional impact force from its initial value of 2π can nevertheless be predicted within the limitations of potential flow theory neglecting gravity and the flexibility of the cylinder; it is clearly related to nonlinear effects.

Conclusion

A rational approach to the problem of the water impact of a circular cylinder was presented. The method justifies the wetting correction introduced by Wagner in 1931, and establishes its domain of validity in this special case. The rapid decrease of the slamming coefficient from its initial value of 2π predicted by the present theory together with the finite rise time observed experimentally, and which cannot be explained within the assumptions made, serve to explain the scatter in the experimental observations concerning the maximum impact force on a cylinder. This is due to the fact that this rise time has to be scaled according to different effects than those described here, and is true even when the flexibility of the cylinder is ignored. Another important conclusion is that one must exercise extreme care in using corrections other than the wetting correction since, for the circular cylinder, different effects lead to corrections of the same order.

The method appears to be applicable to other cases, such as different geometries, nonvertical and/or nonconstant velocities, and eventually curved free surfaces. Such situations, which have not been studied as extensively as the problem addressed here, could lead to interesting new results concerning the different parameters involved in the analysis of the impact loads exerted on a body piercing the free surface due to slamming and/or green seas impact.

The method of matched asymptotic expansions, already used to study the flow past blunt ships [18, 19] and hydroplaning [20, 21], has been proven suitable for dealing with singularities such as the one appearing at the water line for large motions of the partially emerged pontoons of a damaged semi-submersible.

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References

- 1 Dahle, L. A., "Mobile Platforms Stability: Project Synthesis with Recommendations for New Philosophies for Stability Regulations," *Proceedings, 17th Offshore Technology Conference*, OTC Paper 4988, Houston, Tex., 1985.
- 2 Shipbuilding Research Association of Japan (JSRA), Report of SR 192 Research Panel, Research Memoir of the Shipbuilding Research Association of Japan, No. 373, 1984.
- 3 Huang, X., Naess, A., and Hoff, J. R., "On the Behavior of Semisubmersible Platforms at Large List Angles," *Proceedings, 14th Offshore Technology Conference*, OTC Paper 4246, Houston, Tex., 1982.
- 4 Huang, X., and Naess, A., "Dynamic Response of a Heavily Listed Semisubmersible Platform," *Proceedings, Second International Symposium on Ocean Engineering and Ship Handling*, SSPA, Göteborg, Mar. 1983.
- 5 Naess, A., and Hoff, J. R., "Time Simulation of the Dynamic Response of Heavily Listed Semisubmersible Platforms in Waves," NHL, Report 183347, 1984.
- 6 Paulling, J. R., "Time Domain Simulation of Semisubmersible Platform Motion with Application to the Tension Leg Platform," *Proceedings, STAR Symposium*, SNAME, San Francisco, May 1977.
- 7 Lin, W.-M., Newman, J. N., and Yue, D. K., "Nonlinear Forced Motions of Floating Bodies," *Proceedings, 15th Symposium on Naval Hydrodynamics*, Hamburg, Sept. 1984.
- 8 Von Kármán, T., "The Impact of Seaplanes Floats During Landing," NACA TN 321, Oct. 1929.
- 9 Wagner, H., "Landing of Seaplanes," NACA TM 622, 1931.
- 10 Wagner, H., "Über Stoß und Gleitvorgänge an der Oberfläche von Flüssigkeiten," *ZAMM*, Vol. 12, 1932, pp. 193-215.
- 11 Fabula, A., "Ellipse-Fitting Approximation of Two-Dimensional Normal Symmetric Impact of Rigid Bodies on Water," *Proceedings, Fifth Midwestern Conference on Fluid Mechanics*, University of Michigan Press, Ann Arbor, Mich., 1957.
- 12 Schiffman, M., and Spencer, D. C., "The Force of Impact on a Cone Striking a Water Surface (Vertical Entry)," *Comm. Pure Appl. Math.*, Vol. 4, 1951, pp. 379-417.
- 13 Faltinsen, O., "Numerical Solutions of Transient Nonlinear Free-Surface Motion Outside or Inside Moving Bodies," *Proceedings, Second International Conference on Numerical Ship Hydrodynamics*, Berkeley, Calif., 1977, pp. 347-357.
- 14 Faltinsen, O., et al., "Water Impact Loads and Dynamic Responses of Horizontal Circular Cylinders in Offshore Structures," *Proceedings, 9th Offshore Technology Conference*, OTC Paper 2741, Houston, Tex., 1977.
- 15 "Hydrodynamic Impact Analysis," EPRI NP-824, Electric Power Research Institute, Palo Alto, Calif., June 1978.
- 16 Greenhow, M., and Lin, W.-M., "Nonlinear Free Surface Effects: Experiments and Theory," MIT, Ocean Engineering Department, Report No. 83-19, Cambridge, Mass., 1983.
- 17 Taylor, J. L., "Some Hydrodynamical Inertia Coefficients," *Philosophical Magazine*, Ser. 7, Vol. 9, 1930, pp. 161-183.
- 18 Dagan, G., and Tulin, M. P., "Two-Dimensional Free-Surface Gravity Flow Past Blunt Bodies," *Journal of Fluid Mechanics*, Vol. 51, 1972, pp. 529-543.
- 19 Fernandez, G., "Nonlinearity of the Three-Dimensional Flow Past a Flat Blunt Body," *Journal of Fluid Mechanics*, Vol. 108, 1981, pp. 345-361.
- 20 Shen, Y. T., and Ogilvie, T. F., "Nonlinear Hydrodynamic Theory for Finite-Span Planing Surfaces," *Journal of Ship Research*, Mar. 1972, pp. 3-20.
- 21 Nguyen-Ngoc-Tran, "Détermination de l'Épaisseur et de la Direction du Jet dans l'Hydroplanage des Surfaces à Grandes Envergures," *C. R. Acad. Sc. Paris*, Ser. A, Vol. 280, 1975, pp. 463-466.